AIVOA PHARMACEUTICALS MANUFACTURING CORP.
GLOBAL QUALITY OPERATIONS & COMPLIANCE DIVISION

PHARMACEUTICAL API QUALITY DEVIATION & INVESTIGATION REPORT

Deviation Number:	DEV-2026-1002	**Classification:**	Critical / Biological API Purity (CQA)
Product / API Name:	Semaglutide Synthetic Peptide API	**Batch / Lot Number:**	SEM-2026-081
Manufacturing Stage:	Stage 6: Preparative RP-HPLC & Lyophilization	**Equipment Tag:**	Prep-HPLC Skid PREP-LC-03 / Lyophilizer LYO-02
Date of Occurrence:	2026-10-20 18:00 UTC	**Date of Discovery:**	2026-10-22 14:20 UTC
Material / Lot ID:	Ammonium Bicarbonate Lot ABC-4410	**Reporting Unit:**	Biologics Analytical Quality Control

1. OBSERVED DEVIATION & ANALYTICAL TEST RESULTS

Test Parameter	Regulatory Specification	Observed Lab Result	Test Methodology / Instrument
High Molecular Weight Impurities (HMWI / Dimer)	**NMT 0.80% (Area %)**	**1.45% (Area %) [OOS]**	Size Exclusion Chromatography SEC-HPLC-01
Purity by RP-HPLC	**NLT 98.0%**	**96.8%**	Reverse Phase C18 HPLC
Peptide Content	85.0% - 95.0%	88.4%	Kjeldahl Nitrogen Titration
Endotoxin Level	NMT 5.0 EU/mg	0.45 EU/mg	LAL Kinetic Turbidimetric

2. INCIDENT CHRONOLOGY & FORENSIC NARRATIVE

Size Exclusion Chromatography (SEC) analysis of purified Semaglutide peptide batch SEM-2026-081 revealed covalent dimer and multimer formation at 1.45%, exceeding the 0.80% release limit. Buffer preparation records show Mobile Phase B (ammonium bicarbonate buffer) was held at 24°C room temperature for 72 hours instead of validated 24 hours at 2-8°C, leading to CO2 off-gassing, pH shift from 7.5 to 8.7, and accelerated inter-chain disulfide exchange.

3. IMMEDIATE CONTAINMENT ACTIONS TAKEN

1. Lyophilized peptide bulk (3.2 kg) placed under restricted cold quarantine (-20°C Vault).
2. PREP-LC-03 buffer feed manifolds flushed and sanitized.
3. Buffer shelf-life SOP SOP-BUF-041 flagged for urgent revision.

Prepared by:	Dr. L. Zhang (QC Biologics Lead)	2026-10-22
Verified by:	A. Dubois (Downstream Lead)	2026-10-22
QA Approval:	Dr. S. Thornton (QA Lead)	**Status: PENDING ROOT CAUSE ANALYSIS**